

Supplementary Material for “Reliability Failure of Remaining-Useful-Life Prediction Intervals under Sensor Degradation”

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Every table below is generated by script from the named source file in the released repository (jeff-1980/paper2, tag v0.9-review); no value is transcribed by hand. All perturbations are seeded from SHA-256 digests; with PYTHONHASHSEED=0 and deterministic cuDNN, two independent runs on the same machine reproduced the 31 result files listed in REPRODUCE.md bit-for-bit.

S0. PROVENANCE OF EVERY TABLE

Each main-text and appendix table, with the generating script or result file in the released repository, the label version (“clipped” is the 125-cycle target clip), the evaluation unit, whether engines are equally weighted, how seeds and perturbation realisations are aggregated, and the data over which perturbation scales are referenced.

Table	Script / result file	Labels	Unit	Engine-weighted	Aggregation	Noise range
Protocol 2 × 2 (Table I)	mst_step0_retrain_checkpoints.py (T/W), mst_step0b_retrain_valselect.py (V/W), mst_step0c_retrain_leakfree.py (V/F), mst_r2_step7_protocol_2x2.py (T/F)	clipped	terminal	yes	mean of 5 seeds; ECE on pooled seeds	–
Clean benchmark (Table II)	mst_r2_step_fair_main_table.py (fixed variance, MC Dropout); cost-table scripts → table2_clean_full.json (others); per-engine column from mst_step4b_per_engine_coverage_leakfree.py	clipped	terminal (per-engine: trajectory)	yes	mean of 5 seeds; ECE per seed	–
Same-partition ensemble (Table III)	mst_r4_step1_samesplit_ensemble.py	clipped	terminal	yes	one partition (seed 42), five initialisations	–
Attribution, LSTM (Table IV)	mst_r3_stepH_lstm_exact_interp_attribution.py; clamp fraction from mst_step5k_clamp_frac_leakfree.py	clipped	terminal	yes	seed-pooled predictions; interpolated crossover	official training file
Structured degradation (Table V)	mst_r2_engine.py sweeps via mst_r2_partB_degradation.py → r2_degradation_half_life_feat_oob.json	clipped	terminal	yes	5 models × 5 realisations, grand mean	official training file
Drift controls (Table VI)	mst_r2_step5_drift_controls.py; mst_r5_step3_bias_equal_magnitude_control.py; fixed-endpoint shuffle → D1_fixed_endpoint_shuffle.json	clipped	terminal	yes	5 models; deterministic except shuffled/bias sign draws	official training file
Decision stress test (Table VII)	mst_r3_stepC_maintenance_onesided.py; unclipped RUL from mst_r4_step2b_maintenance_rul_correction_deterministic.py	clipped for triggers; unclipped RUL reported	terminal	yes	5 models × 5 realisations	official training file
Fair calibration (Appendix B)	mst_r2_step_fair_main_table.py → fair_calibration_sigma_fixed.json	clipped	terminal	yes	mean of 5 seeds	–
Attribution, Transformer (Appendix C)	mst_r3_stepG_transformer_exact_interp_attribution.py	clipped	terminal	yes	seed-pooled predictions; interpolated crossover	official training file
Seeds and ensemble size (Appendix D)	n5_vs_n15_variance_comparison_leakfree.json; A2_fd004_msweep_IS.json; E_ensemble_independent_replication.json (scripts in code/)	clipped	terminal	yes	15 seeds; three disjoint ensembles	–
Latency (Appendix E)	mst_r2_step4_latency_redesign.py → latency_redesign_run1.json	–	–	–	median and IQR over 11 repetitions	–

S1. FOUR-WAY ATTRIBUTION AT THE INTERPOLATED TEN-POINT-LOSS POINT

Source: leakfree_r3/H_lstm_exact_interp_attribution.json and leakfree_r3/G_transformer_exact_interp_attribution.json. C_{ij} is coverage with mean output from condition i and scale output from condition j (0 clean, 1 perturbed). Order A varies the mean first; order B varies the scale first. Interaction = $(C_{11} - C_{10}) - (C_{01} - C_{00})$, computed from the four coverages; it is positive in all sixteen cells and reaches +0.047.

Cross-seed standard deviations of C_{10} and C_{11} at the grid point nearest the crossover (leakfree_r3/B_attribution_raw_per_seed.json) lie between 0.011 and 0.081; the percentage contributions in the main text are computed from seed-pooled predictions (aggregate-then-difference), not as a mean of per-seed ratios.

Paired bootstrap of mean dominance. For each cell, $|\Delta_\mu| - |\Delta_\sigma|$ was computed per seed at the same interpolated ten-point-loss point as the table above (per-seed C_{ij} interpolated between the same bracketing levels), separately for each decomposition order, and bootstrapped over the five seeds (2000 resamples). The intervals reflect variability across the five training-and-partition seeds only; they are conditional on the fixed test set, the seeded perturbation realisations and the chosen interpolated crossover, and do not propagate those sources of uncertainty. Source: leakfree_r3/bootstrap_by_order_exact_interp.json.

TABLE II
FOUR-WAY ATTRIBUTION, BOTH BACKBONES, AT THE INTERPOLATED POINT (f_{oob} IN %).

Backbone	Data	Mech.	f_{oob}	C_{00}	C_{10}	C_{01}	C_{11}	μ (A)	σ (A)	μ (B)	σ (B)	Int.	Both
LSTM	FD001	Heterosc.	0.74	0.864	0.794	0.828	0.764	-0.0700	-0.0300	-0.0642	-0.0358	+0.0058	yes
LSTM	FD001	CP-norm	0.62	0.894	0.813	0.852	0.794	-0.0808	-0.0192	-0.0578	-0.0422	+0.0230	yes
LSTM	FD002	Heterosc.	1.43	0.918	0.827	0.888	0.818	-0.0908	-0.0092	-0.0703	-0.0297	+0.0206	yes
LSTM	FD002	CP-norm	1.39	0.934	0.847	0.902	0.834	-0.0877	-0.0123	-0.0680	-0.0320	+0.0196	yes
LSTM	FD003	Heterosc.	0.31	0.858	0.760	0.837	0.758	-0.0979	-0.0020	-0.0790	-0.0210	+0.0190	yes
LSTM	FD003	CP-norm	0.48	0.910	0.818	0.877	0.810	-0.0924	-0.0076	-0.0667	-0.0333	+0.0256	yes
LSTM	FD004	Heterosc.	1.20	0.892	0.781	0.856	0.792	-0.1108	+0.0108	-0.0636	-0.0364	+0.0472	yes
LSTM	FD004	CP-norm	1.16	0.889	0.781	0.852	0.789	-0.1087	+0.0087	-0.0630	-0.0370	+0.0457	yes
Transformer	FD001	Heterosc.	0.86	0.886	0.803	0.865	0.786	-0.0830	-0.0170	-0.0788	-0.0212	+0.0042	yes
Transformer	FD001	CP-norm	0.65	0.894	0.808	0.868	0.794	-0.0859	-0.0141	-0.0737	-0.0263	+0.0121	yes
Transformer	FD002	Heterosc.	1.33	0.884	0.797	0.855	0.784	-0.0873	-0.0127	-0.0716	-0.0284	+0.0157	yes
Transformer	FD002	CP-norm	1.34	0.903	0.814	0.871	0.803	-0.0887	-0.0113	-0.0684	-0.0316	+0.0204	yes
Transformer	FD003	Heterosc.	0.12	0.870	0.770	0.850	0.770	-0.0999	-0.0001	-0.0797	-0.0203	+0.0203	yes
Transformer	FD003	CP-norm	0.21	0.886	0.791	0.865	0.786	-0.0948	-0.0052	-0.0790	-0.0210	+0.0158	yes
Transformer	FD004	Heterosc.	0.71	0.857	0.759	0.823	0.757	-0.0973	-0.0027	-0.0664	-0.0336	+0.0309	yes
Transformer	FD004	CP-norm	0.73	0.856	0.762	0.827	0.756	-0.0937	-0.0063	-0.0712	-0.0288	+0.0225	yes

Backbone	Data	Mech.	Mean first: mean [95% CI]	> 0	Scale first: mean [95% CI]	> 0
LSTM	FD001	Heterosc.	+0.040 [+0.021, +0.059]	yes	+0.028 [+0.015, +0.041]	yes
LSTM	FD001	CP-norm	+0.062 [+0.042, +0.083]	yes	+0.016 [-0.008, +0.038]	no
LSTM	FD002	Heterosc.	+0.081 [+0.071, +0.091]	yes	+0.041 [+0.029, +0.051]	yes
LSTM	FD002	CP-norm	+0.073 [+0.051, +0.097]	yes	+0.036 [+0.022, +0.049]	yes
LSTM	FD003	Heterosc.	+0.086 [+0.061, +0.111]	yes	+0.052 [+0.042, +0.063]	yes
LSTM	FD003	CP-norm	+0.079 [+0.040, +0.124]	yes	+0.033 [+0.026, +0.045]	yes
LSTM	FD004	Heterosc.	+0.100 [+0.092, +0.108]	yes	+0.027 [+0.013, +0.040]	yes
LSTM	FD004	CP-norm	+0.100 [+0.089, +0.113]	yes	+0.026 [+0.022, +0.030]	yes
Transformer	FD001	Heterosc.	+0.066 [+0.038, +0.094]	yes	+0.058 [+0.054, +0.060]	yes
Transformer	FD001	CP-norm	+0.072 [+0.047, +0.103]	yes	+0.047 [+0.038, +0.058]	yes
Transformer	FD002	Heterosc.	+0.075 [+0.062, +0.088]	yes	+0.043 [+0.035, +0.051]	yes
Transformer	FD002	CP-norm	+0.077 [+0.060, +0.095]	yes	+0.037 [+0.029, +0.044]	yes
Transformer	FD003	Heterosc.	+0.090 [+0.061, +0.118]	yes	+0.058 [+0.045, +0.073]	yes
Transformer	FD003	CP-norm	+0.085 [+0.041, +0.130]	yes	+0.058 [+0.051, +0.065]	yes
Transformer	FD004	Heterosc.	+0.092 [+0.079, +0.107]	yes	+0.033 [+0.026, +0.040]	yes
Transformer	FD004	CP-norm	+0.087 [+0.075, +0.100]	yes	+0.042 [+0.036, +0.049]	yes

S2. TERMINAL-INSTANT DECISION STRESS TEST, FULL RESULTS

Source: leakfree_r3/C_maintenance_full_onesided.json. One-sided 95% lower bound; unrec. = unrecognised maintenance need (overall / conditional on true $RUL \leq L$); prem. = premature trigger rate; $RUL@t$ = true RUL at premature trigger under the 125-cycle label clip (unclipped values for the main-text rows are in Table VII of the paper); cost = $c_{\text{late}}r_{\text{late}} + c_{\text{early}}r_{\text{early}}$ with $c_{\text{early}} = 1$; rank at ratios 5/20/100.

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	$RUL@t$	cost 5/20/100	rank
LSTM	FD001	Clean	10	Heterosc.	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Clean	10	Fixed var.	0.0000 / 0.000	0.0180	38.4	0.018 / 0.018 / 0.018	4/4/4
LSTM	FD001	Clean	10	MC Drop.	0.0000 / 0.000	0.0180	38.4	0.018 / 0.018 / 0.018	5/5/5
LSTM	FD001	Clean	10	Ensemble	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	2/2/2
LSTM	FD001	Clean	10	CP-norm	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	3/3/3
LSTM	FD001	Clean	20	Heterosc.	0.0020 / 0.013	0.0000	—	0.010 / 0.040 / 0.200	3/5/5
LSTM	FD001	Clean	20	Fixed var.	0.0000 / 0.000	0.0220	55.6	0.022 / 0.022 / 0.022	4/3/3
LSTM	FD001	Clean	20	MC Drop.	0.0000 / 0.000	0.0220	55.6	0.022 / 0.022 / 0.022	5/4/4
LSTM	FD001	Clean	20	Ensemble	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Clean	20	CP-norm	0.0000 / 0.000	0.0020	47.0	0.002 / 0.002 / 0.002	2/2/2
LSTM	FD001	Clean	30	Heterosc.	0.0000 / 0.000	0.0080	61.5	0.008 / 0.008 / 0.008	2/2/2
LSTM	FD001	Clean	30	Fixed var.	0.0000 / 0.000	0.0300	69.3	0.030 / 0.030 / 0.030	4/4/4
LSTM	FD001	Clean	30	MC Drop.	0.0000 / 0.000	0.0300	69.3	0.030 / 0.030 / 0.030	5/5/5
LSTM	FD001	Clean	30	Ensemble	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Clean	30	CP-norm	0.0000 / 0.000	0.0220	70.7	0.022 / 0.022 / 0.022	3/3/3
LSTM	FD001	Noise 1%	10	Heterosc.	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0172	37.7	0.017 / 0.017 / 0.017	4/4/4
LSTM	FD001	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0180	38.4	0.018 / 0.018 / 0.018	5/5/5
LSTM	FD001	Noise 1%	10	Ensemble	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	2/2/2
LSTM	FD001	Noise 1%	10	CP-norm	0.0000 / 0.000	0.0004	34.0	0.000 / 0.000 / 0.000	3/3/3
LSTM	FD001	Noise 1%	20	Heterosc.	0.0020 / 0.013	0.0000	—	0.010 / 0.040 / 0.200	3/5/5

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
LSTM	FD001	Noise 1%	20	Fixed var.	0.0000 / 0.000	0.0208	55.3	0.021 / 0.021 / 0.021	4/3/2
LSTM	FD001	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0232	56.8	0.023 / 0.023 / 0.023	5/4/3
LSTM	FD001	Noise 1%	20	Ensemble	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Noise 1%	20	CP-norm	0.0004 / 0.003	0.0020	47.0	0.004 / 0.010 / 0.042	2/2/4
LSTM	FD001	Noise 1%	30	Heterosc.	0.0008 / 0.003	0.0072	63.2	0.011 / 0.023 / 0.087	2/3/5
LSTM	FD001	Noise 1%	30	Fixed var.	0.0000 / 0.000	0.0296	69.1	0.030 / 0.030 / 0.030	4/4/3
LSTM	FD001	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0300	69.3	0.030 / 0.030 / 0.030	5/5/4
LSTM	FD001	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0020	69.0	0.002 / 0.002 / 0.002	1/1/1
LSTM	FD001	Noise 1%	30	CP-norm	0.0000 / 0.000	0.0188	66.8	0.019 / 0.019 / 0.019	3/2/2
LSTM	FD001	Bias 5%	10	Heterosc.	0.0000 / 0.000	0.0028	38.0	0.003 / 0.003 / 0.003	1/1/1
LSTM	FD001	Bias 5%	10	Fixed var.	0.0000 / 0.000	0.0188	39.4	0.019 / 0.019 / 0.019	4/4/4
LSTM	FD001	Bias 5%	10	MC Drop.	0.0000 / 0.000	0.0192	40.0	0.019 / 0.019 / 0.019	5/5/5
LSTM	FD001	Bias 5%	10	Ensemble	0.0000 / 0.000	0.0040	38.0	0.004 / 0.004 / 0.004	2/2/2
LSTM	FD001	Bias 5%	10	CP-norm	0.0000 / 0.000	0.0056	40.5	0.006 / 0.006 / 0.006	3/3/3
LSTM	FD001	Bias 5%	20	Heterosc.	0.0028 / 0.018	0.0008	58.0	0.015 / 0.057 / 0.281	2/5/5
LSTM	FD001	Bias 5%	20	Fixed var.	0.0008 / 0.005	0.0264	60.2	0.030 / 0.042 / 0.106	5/4/3
LSTM	FD001	Bias 5%	20	MC Drop.	0.0004 / 0.003	0.0272	59.9	0.029 / 0.035 / 0.067	4/2/2
LSTM	FD001	Bias 5%	20	Ensemble	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Bias 5%	20	CP-norm	0.0012 / 0.007	0.0120	76.0	0.018 / 0.036 / 0.132	3/3/4
LSTM	FD001	Bias 5%	30	Heterosc.	0.0032 / 0.013	0.0144	69.4	0.030 / 0.078 / 0.334	2/5/5
LSTM	FD001	Bias 5%	30	Fixed var.	0.0012 / 0.005	0.0316	70.1	0.038 / 0.056 / 0.152	3/2/2
LSTM	FD001	Bias 5%	30	MC Drop.	0.0012 / 0.005	0.0336	70.3	0.040 / 0.058 / 0.154	4/3/3
LSTM	FD001	Bias 5%	30	Ensemble	0.0000 / 0.000	0.0220	69.1	0.022 / 0.022 / 0.022	1/1/1
LSTM	FD001	Bias 5%	30	CP-norm	0.0016 / 0.006	0.0384	78.3	0.046 / 0.070 / 0.198	5/4/4
LSTM	FD001	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.0180	36.2	0.018 / 0.018 / 0.018	1/1/1
LSTM	FD001	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.0720	49.4	0.072 / 0.072 / 0.072	4/4/4
LSTM	FD001	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.0740	50.4	0.074 / 0.074 / 0.074	5/5/5
LSTM	FD001	Drift 5%	10	Ensemble	0.0000 / 0.000	0.0200	36.0	0.020 / 0.020 / 0.020	2/2/2
LSTM	FD001	Drift 5%	10	CP-norm	0.0000 / 0.000	0.0280	39.3	0.028 / 0.028 / 0.028	3/3/3
LSTM	FD001	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.0260	63.0	0.026 / 0.026 / 0.026	1/1/1
LSTM	FD001	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.1180	62.5	0.118 / 0.118 / 0.118	4/4/4
LSTM	FD001	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.1200	62.3	0.120 / 0.120 / 0.120	5/5/5
LSTM	FD001	Drift 5%	20	Ensemble	0.0000 / 0.000	0.0300	55.3	0.030 / 0.030 / 0.030	2/2/2
LSTM	FD001	Drift 5%	20	CP-norm	0.0000 / 0.000	0.0580	67.7	0.058 / 0.058 / 0.058	3/3/3
LSTM	FD001	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.0680	69.3	0.068 / 0.068 / 0.068	1/1/1
LSTM	FD001	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.1360	78.8	0.136 / 0.136 / 0.136	3/3/3
LSTM	FD001	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.1400	78.6	0.140 / 0.140 / 0.140	4/4/4
LSTM	FD001	Drift 5%	30	Ensemble	0.0000 / 0.000	0.1600	79.1	0.160 / 0.160 / 0.160	5/5/5
LSTM	FD001	Drift 5%	30	CP-norm	0.0000 / 0.000	0.1340	81.9	0.134 / 0.134 / 0.134	2/2/2
LSTM	FD002	Clean	10	Heterosc.	0.0000 / 0.000	0.0032	32.5	0.003 / 0.003 / 0.003	1/1/1
LSTM	FD002	Clean	10	Fixed var.	0.0000 / 0.000	0.0656	42.7	0.066 / 0.066 / 0.066	4/4/4
LSTM	FD002	Clean	10	MC Drop.	0.0000 / 0.000	0.0664	42.6	0.066 / 0.066 / 0.066	5/5/5
LSTM	FD002	Clean	10	Ensemble	0.0000 / 0.000	0.0079	37.0	0.008 / 0.008 / 0.008	2/2/2
LSTM	FD002	Clean	10	CP-norm	0.0000 / 0.000	0.0119	46.1	0.012 / 0.012 / 0.012	3/3/3
LSTM	FD002	Clean	20	Heterosc.	0.0000 / 0.000	0.0372	50.4	0.037 / 0.037 / 0.037	1/1/1
LSTM	FD002	Clean	20	Fixed var.	0.0000 / 0.000	0.0783	53.6	0.078 / 0.078 / 0.078	4/4/4
LSTM	FD002	Clean	20	MC Drop.	0.0000 / 0.000	0.0814	54.0	0.081 / 0.081 / 0.081	5/5/5
LSTM	FD002	Clean	20	Ensemble	0.0000 / 0.000	0.0514	49.2	0.051 / 0.051 / 0.051	2/2/2
LSTM	FD002	Clean	20	CP-norm	0.0000 / 0.000	0.0538	54.9	0.054 / 0.054 / 0.054	3/3/3
LSTM	FD002	Clean	30	Heterosc.	0.0000 / 0.000	0.0743	69.0	0.074 / 0.074 / 0.074	3/3/3
LSTM	FD002	Clean	30	Fixed var.	0.0000 / 0.000	0.0719	65.0	0.072 / 0.072 / 0.072	1/1/1
LSTM	FD002	Clean	30	MC Drop.	0.0000 / 0.000	0.0719	65.2	0.072 / 0.072 / 0.072	2/2/2
LSTM	FD002	Clean	30	Ensemble	0.0000 / 0.000	0.0870	68.1	0.087 / 0.087 / 0.087	5/5/5
LSTM	FD002	Clean	30	CP-norm	0.0000 / 0.000	0.0846	70.6	0.085 / 0.085 / 0.085	4/4/4
LSTM	FD002	Noise 1%	10	Heterosc.	0.0021 / 0.024	0.0090	41.4	0.019 / 0.050 / 0.215	1/2/4
LSTM	FD002	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0783	46.9	0.078 / 0.078 / 0.078	4/4/1
LSTM	FD002	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0813	47.1	0.081 / 0.081 / 0.081	5/5/2
LSTM	FD002	Noise 1%	10	Ensemble	0.0016 / 0.018	0.0166	39.2	0.025 / 0.048 / 0.175	2/1/3
LSTM	FD002	Noise 1%	10	CP-norm	0.0024 / 0.027	0.0168	47.9	0.029 / 0.064 / 0.254	3/3/5
LSTM	FD002	Noise 1%	20	Heterosc.	0.0024 / 0.011	0.0506	58.0	0.062 / 0.098 / 0.288	1/4/5
LSTM	FD002	Noise 1%	20	Fixed var.	0.0003 / 0.002	0.0868	57.9	0.088 / 0.093 / 0.118	4/2/1
LSTM	FD002	Noise 1%	20	MC Drop.	0.0003 / 0.002	0.0884	58.0	0.090 / 0.095 / 0.120	5/3/2
LSTM	FD002	Noise 1%	20	Ensemble	0.0008 / 0.004	0.0688	57.9	0.073 / 0.085 / 0.148	2/1/3
LSTM	FD002	Noise 1%	20	CP-norm	0.0019 / 0.009	0.0672	63.3	0.077 / 0.105 / 0.257	3/5/4
LSTM	FD002	Noise 1%	30	Heterosc.	0.0003 / 0.001	0.0933	78.3	0.095 / 0.100 / 0.125	3/3/5
LSTM	FD002	Noise 1%	30	Fixed var.	0.0002 / 0.001	0.0833	72.8	0.084 / 0.086 / 0.099	1/1/1
LSTM	FD002	Noise 1%	30	MC Drop.	0.0002 / 0.001	0.0851	72.9	0.086 / 0.088 / 0.101	2/2/2
LSTM	FD002	Noise 1%	30	Ensemble	0.0000 / 0.000	0.1083	78.5	0.108 / 0.108 / 0.108	5/5/3
LSTM	FD002	Noise 1%	30	CP-norm	0.0002 / 0.001	0.1031	79.4	0.104 / 0.106 / 0.119	4/4/4
LSTM	FD002	Bias 5%	10	Heterosc.	0.0397 / 0.456	0.1772	89.8	0.376 / 0.971 / 4.146	5/5/5
LSTM	FD002	Bias 5%	10	Fixed var.	0.0286 / 0.329	0.1564	90.1	0.299 / 0.729 / 3.018	3/3/3
LSTM	FD002	Bias 5%	10	MC Drop.	0.0280 / 0.322	0.1565	90.0	0.296 / 0.716 / 2.955	2/2/2
LSTM	FD002	Bias 5%	10	Ensemble	0.0158 / 0.182	0.1818	86.4	0.261 / 0.498 / 1.763	1/1/1
LSTM	FD002	Bias 5%	10	CP-norm	0.0373 / 0.429	0.1764	89.8	0.363 / 0.923 / 3.908	4/4/4
LSTM	FD002	Bias 5%	20	Heterosc.	0.0729 / 0.348	0.1668	94.1	0.531 / 1.625 / 7.455	5/5/5

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
LSTM	FD002	Bias 5%	20	Fixed var.	0.0713 / 0.340	0.1459	94.2	0.502 / 1.572 / 7.276	4/4/4
LSTM	FD002	Bias 5%	20	MC Drop.	0.0710 / 0.339	0.1464	94.2	0.501 / 1.566 / 7.245	3/3/3
LSTM	FD002	Bias 5%	20	Ensemble	0.0292 / 0.140	0.1881	91.8	0.334 / 0.773 / 3.113	1/1/1
LSTM	FD002	Bias 5%	20	CP-norm	0.0643 / 0.307	0.1677	93.8	0.489 / 1.455 / 6.603	2/2/2
LSTM	FD002	Bias 5%	30	Heterosc.	0.0585 / 0.243	0.1573	98.4	0.450 / 1.327 / 6.007	3/3/3
LSTM	FD002	Bias 5%	30	Fixed var.	0.0781 / 0.324	0.1361	98.3	0.527 / 1.698 / 7.946	5/5/5
LSTM	FD002	Bias 5%	30	MC Drop.	0.0768 / 0.319	0.1364	98.3	0.521 / 1.673 / 7.820	4/4/4
LSTM	FD002	Bias 5%	30	Ensemble	0.0285 / 0.118	0.1802	97.4	0.323 / 0.749 / 3.026	1/1/1
LSTM	FD002	Bias 5%	30	CP-norm	0.0568 / 0.235	0.1575	98.1	0.441 / 1.293 / 5.833	2/2/2
LSTM	FD002	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.7233	90.0	0.723 / 0.723 / 0.723	2/2/2
LSTM	FD002	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.7557	91.2	0.756 / 0.756 / 0.756	4/4/4
LSTM	FD002	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.7557	91.2	0.756 / 0.756 / 0.756	5/5/5
LSTM	FD002	Drift 5%	10	Ensemble	0.0000 / 0.000	0.7431	90.7	0.743 / 0.743 / 0.743	3/3/3
LSTM	FD002	Drift 5%	10	CP-norm	0.0000 / 0.000	0.7194	89.9	0.719 / 0.719 / 0.719	1/1/1
LSTM	FD002	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.6996	95.6	0.700 / 0.700 / 0.700	2/2/2
LSTM	FD002	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.7020	95.7	0.702 / 0.702 / 0.702	4/4/4
LSTM	FD002	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.7020	95.7	0.702 / 0.702 / 0.702	5/5/5
LSTM	FD002	Drift 5%	20	Ensemble	0.0000 / 0.000	0.6996	95.6	0.700 / 0.700 / 0.700	3/3/3
LSTM	FD002	Drift 5%	20	CP-norm	0.0000 / 0.000	0.6980	95.5	0.698 / 0.698 / 0.698	1/1/1
LSTM	FD002	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.6514	99.7	0.651 / 0.651 / 0.651	1/1/1
LSTM	FD002	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.6522	99.7	0.652 / 0.652 / 0.652	3/3/3
LSTM	FD002	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.6522	99.7	0.652 / 0.652 / 0.652	4/4/4
LSTM	FD002	Drift 5%	30	Ensemble	0.0000 / 0.000	0.6522	99.7	0.652 / 0.652 / 0.652	5/5/5
LSTM	FD002	Drift 5%	30	CP-norm	0.0000 / 0.000	0.6514	99.7	0.651 / 0.651 / 0.651	2/2/2
LSTM	FD003	Clean	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD003	Clean	10	Fixed var.	0.0000 / 0.000	0.0260	42.2	0.026 / 0.026 / 0.026	5/5/5
LSTM	FD003	Clean	10	MC Drop.	0.0000 / 0.000	0.0240	39.8	0.024 / 0.024 / 0.024	4/4/4
LSTM	FD003	Clean	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
LSTM	FD003	Clean	10	CP-norm	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	3/3/3
LSTM	FD003	Clean	20	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD003	Clean	20	Fixed var.	0.0000 / 0.000	0.0460	53.0	0.046 / 0.046 / 0.046	4/4/4
LSTM	FD003	Clean	20	MC Drop.	0.0000 / 0.000	0.0500	52.7	0.050 / 0.050 / 0.050	5/5/5
LSTM	FD003	Clean	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
LSTM	FD003	Clean	20	CP-norm	0.0000 / 0.000	0.0320	64.6	0.032 / 0.032 / 0.032	3/3/3
LSTM	FD003	Clean	30	Heterosc.	0.0000 / 0.000	0.0140	61.9	0.014 / 0.014 / 0.014	1/1/1
LSTM	FD003	Clean	30	Fixed var.	0.0000 / 0.000	0.0460	60.0	0.046 / 0.046 / 0.046	3/3/3
LSTM	FD003	Clean	30	MC Drop.	0.0000 / 0.000	0.0540	59.4	0.054 / 0.054 / 0.054	4/4/4
LSTM	FD003	Clean	30	Ensemble	0.0000 / 0.000	0.0300	60.3	0.030 / 0.030 / 0.030	2/2/2
LSTM	FD003	Clean	30	CP-norm	0.0000 / 0.000	0.0740	64.2	0.074 / 0.074 / 0.074	5/5/5
LSTM	FD003	Noise 1%	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD003	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0240	41.1	0.024 / 0.024 / 0.024	5/5/5
LSTM	FD003	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0236	41.3	0.024 / 0.024 / 0.024	4/4/4
LSTM	FD003	Noise 1%	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
LSTM	FD003	Noise 1%	10	CP-norm	0.0000 / 0.000	0.0012	69.0	0.001 / 0.001 / 0.001	3/3/3
LSTM	FD003	Noise 1%	20	Heterosc.	0.0000 / 0.000	0.0008	55.0	0.001 / 0.001 / 0.001	2/2/2
LSTM	FD003	Noise 1%	20	Fixed var.	0.0000 / 0.000	0.0420	52.9	0.042 / 0.042 / 0.042	4/4/4
LSTM	FD003	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0448	52.2	0.045 / 0.045 / 0.045	5/5/5
LSTM	FD003	Noise 1%	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD003	Noise 1%	20	CP-norm	0.0000 / 0.000	0.0292	64.5	0.029 / 0.029 / 0.029	3/3/3
LSTM	FD003	Noise 1%	30	Heterosc.	0.0000 / 0.000	0.0168	61.3	0.017 / 0.017 / 0.017	1/1/1
LSTM	FD003	Noise 1%	30	Fixed var.	0.0000 / 0.000	0.0520	59.9	0.052 / 0.052 / 0.052	3/3/3
LSTM	FD003	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0564	59.6	0.056 / 0.056 / 0.056	4/4/4
LSTM	FD003	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0340	59.7	0.034 / 0.034 / 0.034	2/2/2
LSTM	FD003	Noise 1%	30	CP-norm	0.0000 / 0.000	0.0820	64.6	0.082 / 0.082 / 0.082	5/5/5
LSTM	FD003	Bias 5%	10	Heterosc.	0.0000 / 0.000	0.0040	35.0	0.004 / 0.004 / 0.004	1/1/1
LSTM	FD003	Bias 5%	10	Fixed var.	0.0000 / 0.000	0.0380	44.0	0.038 / 0.038 / 0.038	4/4/4
LSTM	FD003	Bias 5%	10	MC Drop.	0.0000 / 0.000	0.0412	44.9	0.041 / 0.041 / 0.041	5/5/5
LSTM	FD003	Bias 5%	10	Ensemble	0.0000 / 0.000	0.0040	33.0	0.004 / 0.004 / 0.004	2/2/2
LSTM	FD003	Bias 5%	10	CP-norm	0.0000 / 0.000	0.0080	42.5	0.008 / 0.008 / 0.008	3/3/3
LSTM	FD003	Bias 5%	20	Heterosc.	0.0008 / 0.006	0.0068	54.4	0.011 / 0.023 / 0.087	2/2/3
LSTM	FD003	Bias 5%	20	Fixed var.	0.0008 / 0.006	0.0700	53.0	0.074 / 0.086 / 0.150	4/4/4
LSTM	FD003	Bias 5%	20	MC Drop.	0.0008 / 0.006	0.0724	53.1	0.076 / 0.088 / 0.152	5/5/5
LSTM	FD003	Bias 5%	20	Ensemble	0.0000 / 0.000	0.0060	58.3	0.006 / 0.006 / 0.006	1/1/1
LSTM	FD003	Bias 5%	20	CP-norm	0.0000 / 0.000	0.0496	63.1	0.050 / 0.050 / 0.050	3/3/2
LSTM	FD003	Bias 5%	30	Heterosc.	0.0012 / 0.006	0.0420	59.7	0.048 / 0.066 / 0.162	1/1/4
LSTM	FD003	Bias 5%	30	Fixed var.	0.0008 / 0.004	0.0752	62.5	0.079 / 0.091 / 0.155	3/4/3
LSTM	FD003	Bias 5%	30	MC Drop.	0.0004 / 0.002	0.0792	62.4	0.081 / 0.087 / 0.119	4/3/2
LSTM	FD003	Bias 5%	30	Ensemble	0.0000 / 0.000	0.0700	60.5	0.070 / 0.070 / 0.070	2/2/1
LSTM	FD003	Bias 5%	30	CP-norm	0.0020 / 0.010	0.1084	67.5	0.118 / 0.148 / 0.308	5/5/5
LSTM	FD003	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.0320	42.9	0.032 / 0.032 / 0.032	2/2/2
LSTM	FD003	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.2460	56.2	0.246 / 0.246 / 0.246	4/4/4
LSTM	FD003	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.2500	56.2	0.250 / 0.250 / 0.250	5/5/5
LSTM	FD003	Drift 5%	10	Ensemble	0.0000 / 0.000	0.0300	45.7	0.030 / 0.030 / 0.030	1/1/1
LSTM	FD003	Drift 5%	10	CP-norm	0.0000 / 0.000	0.0540	44.1	0.054 / 0.054 / 0.054	3/3/3
LSTM	FD003	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.1540	54.5	0.154 / 0.154 / 0.154	1/1/1

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
LSTM	FD003	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.3740	68.2	0.374 / 0.374 / 0.374	4/4/4
LSTM	FD003	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.3800	68.6	0.380 / 0.380 / 0.380	5/5/5
LSTM	FD003	Drift 5%	20	Ensemble	0.0000 / 0.000	0.1800	55.1	0.180 / 0.180 / 0.180	2/2/2
LSTM	FD003	Drift 5%	30	CP-norm	0.0000 / 0.000	0.2060	59.8	0.206 / 0.206 / 0.206	3/3/3
LSTM	FD003	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.2700	70.7	0.270 / 0.270 / 0.270	1/1/1
LSTM	FD003	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.4420	82.0	0.442 / 0.442 / 0.442	4/4/4
LSTM	FD003	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.4480	82.1	0.448 / 0.448 / 0.448	5/5/5
LSTM	FD003	Drift 5%	30	Ensemble	0.0000 / 0.000	0.2800	70.1	0.280 / 0.280 / 0.280	2/2/2
LSTM	FD003	Drift 5%	30	CP-norm	0.0000 / 0.000	0.3740	80.1	0.374 / 0.374 / 0.374	3/3/3
LSTM	FD004	Clean	10	Heterosc.	0.0034 / 0.057	0.0371	42.2	0.054 / 0.105 / 0.375	1/5/5
LSTM	FD004	Clean	10	Fixed var.	0.0000 / 0.000	0.0768	43.0	0.077 / 0.077 / 0.077	4/3/3
LSTM	FD004	Clean	10	MC Drop.	0.0000 / 0.000	0.0776	43.1	0.078 / 0.078 / 0.078	5/4/4
LSTM	FD004	Clean	10	Ensemble	0.0000 / 0.000	0.0549	43.8	0.055 / 0.055 / 0.055	2/1/1
LSTM	FD004	Clean	10	CP-norm	0.0000 / 0.000	0.0692	47.8	0.069 / 0.069 / 0.069	3/2/2
LSTM	FD004	Clean	20	Heterosc.	0.0008 / 0.005	0.0473	57.4	0.051 / 0.064 / 0.132	1/2/3
LSTM	FD004	Clean	20	Fixed var.	0.0017 / 0.010	0.0430	55.4	0.051 / 0.077 / 0.212	2/4/4
LSTM	FD004	Clean	20	MC Drop.	0.0017 / 0.010	0.0456	55.4	0.054 / 0.079 / 0.214	3/5/5
LSTM	FD004	Clean	20	Ensemble	0.0000 / 0.000	0.0549	58.4	0.055 / 0.055 / 0.055	4/1/1
LSTM	FD004	Clean	20	CP-norm	0.0000 / 0.000	0.0759	67.4	0.076 / 0.076 / 0.076	5/3/2
LSTM	FD004	Clean	30	Heterosc.	0.0025 / 0.011	0.0726	77.0	0.085 / 0.123 / 0.326	3/1/2
LSTM	FD004	Clean	30	Fixed var.	0.0042 / 0.019	0.0414	71.3	0.062 / 0.126 / 0.463	1/2/3
LSTM	FD004	Clean	30	MC Drop.	0.0042 / 0.019	0.0422	71.2	0.063 / 0.127 / 0.464	2/3/4
LSTM	FD004	Clean	30	Ensemble	0.0042 / 0.019	0.0717	74.5	0.093 / 0.156 / 0.494	4/5/5
LSTM	FD004	Clean	30	CP-norm	0.0008 / 0.004	0.1097	80.4	0.114 / 0.127 / 0.194	5/4/1
LSTM	FD004	Noise 1%	10	Heterosc.	0.0054 / 0.091	0.0495	47.6	0.076 / 0.157 / 0.590	2/5/5
LSTM	FD004	Noise 1%	10	Fixed var.	0.0008 / 0.014	0.0778	45.5	0.082 / 0.095 / 0.162	3/2/2
LSTM	FD004	Noise 1%	10	MC Drop.	0.0007 / 0.011	0.0802	45.9	0.084 / 0.094 / 0.148	4/1/1
LSTM	FD004	Noise 1%	10	Ensemble	0.0034 / 0.057	0.0557	45.7	0.073 / 0.123 / 0.393	1/3/4
LSTM	FD004	Noise 1%	10	CP-norm	0.0025 / 0.043	0.0861	55.5	0.099 / 0.137 / 0.339	5/4/3
LSTM	FD004	Noise 1%	20	Heterosc.	0.0029 / 0.017	0.0694	66.0	0.084 / 0.127 / 0.356	3/4/5
LSTM	FD004	Noise 1%	20	Fixed var.	0.0027 / 0.016	0.0626	61.3	0.076 / 0.117 / 0.333	1/3/4
LSTM	FD004	Noise 1%	20	MC Drop.	0.0024 / 0.014	0.0643	61.8	0.076 / 0.112 / 0.301	2/1/3
LSTM	FD004	Noise 1%	20	Ensemble	0.0017 / 0.010	0.0810	64.7	0.089 / 0.115 / 0.250	4/2/1
LSTM	FD004	Noise 1%	20	CP-norm	0.0015 / 0.009	0.1041	72.0	0.112 / 0.135 / 0.256	5/5/2
LSTM	FD004	Noise 1%	30	Heterosc.	0.0029 / 0.013	0.0965	81.8	0.111 / 0.154 / 0.383	4/3/3
LSTM	FD004	Noise 1%	30	Fixed var.	0.0054 / 0.024	0.0657	78.5	0.093 / 0.174 / 0.606	1/4/4
LSTM	FD004	Noise 1%	30	MC Drop.	0.0056 / 0.025	0.0682	78.8	0.096 / 0.180 / 0.625	2/5/5
LSTM	FD004	Noise 1%	30	Ensemble	0.0008 / 0.004	0.1021	81.4	0.106 / 0.119 / 0.186	3/1/1
LSTM	FD004	Noise 1%	30	CP-norm	0.0010 / 0.005	0.1306	84.5	0.136 / 0.151 / 0.232	5/2/2
LSTM	FD004	Bias 5%	10	Heterosc.	0.0157 / 0.266	0.2442	86.5	0.323 / 0.558 / 1.814	1/3/3
LSTM	FD004	Bias 5%	10	Fixed var.	0.0155 / 0.263	0.2866	86.6	0.364 / 0.597 / 1.839	4/4/4
LSTM	FD004	Bias 5%	10	MC Drop.	0.0157 / 0.266	0.2879	86.7	0.366 / 0.602 / 1.858	5/5/5
LSTM	FD004	Bias 5%	10	Ensemble	0.0118 / 0.200	0.2920	89.7	0.351 / 0.528 / 1.473	3/1/1
LSTM	FD004	Bias 5%	10	CP-norm	0.0133 / 0.226	0.2643	87.1	0.331 / 0.531 / 1.598	2/2/2
LSTM	FD004	Bias 5%	20	Heterosc.	0.0419 / 0.248	0.2488	94.6	0.458 / 1.086 / 4.434	3/4/4
LSTM	FD004	Bias 5%	20	Fixed var.	0.0417 / 0.247	0.2707	95.0	0.479 / 1.104 / 4.439	5/5/5
LSTM	FD004	Bias 5%	20	MC Drop.	0.0407 / 0.241	0.2714	94.9	0.475 / 1.085 / 4.339	4/3/3
LSTM	FD004	Bias 5%	20	Ensemble	0.0329 / 0.195	0.2785	95.7	0.443 / 0.937 / 3.570	1/1/1
LSTM	FD004	Bias 5%	20	CP-norm	0.0380 / 0.225	0.2621	94.7	0.452 / 1.022 / 4.060	2/2/2
LSTM	FD004	Bias 5%	30	Heterosc.	0.0528 / 0.236	0.2543	99.6	0.518 / 1.311 / 5.537	3/5/5
LSTM	FD004	Bias 5%	30	Fixed var.	0.0523 / 0.234	0.2618	99.8	0.523 / 1.308 / 5.494	5/4/4
LSTM	FD004	Bias 5%	30	MC Drop.	0.0518 / 0.232	0.2633	99.8	0.522 / 1.300 / 5.445	4/3/3
LSTM	FD004	Bias 5%	30	Ensemble	0.0430 / 0.192	0.2768	100.3	0.492 / 1.138 / 4.581	1/1/1
LSTM	FD004	Bias 5%	30	CP-norm	0.0481 / 0.215	0.2646	99.9	0.505 / 1.227 / 5.075	2/2/2
LSTM	FD004	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.6726	90.5	0.673 / 0.673 / 0.673	1/1/1
LSTM	FD004	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.7401	91.8	0.740 / 0.740 / 0.740	3/3/3
LSTM	FD004	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.7409	91.9	0.741 / 0.741 / 0.741	4/4/4
LSTM	FD004	Drift 5%	10	Ensemble	0.0000 / 0.000	0.7595	92.6	0.759 / 0.759 / 0.759	5/5/5
LSTM	FD004	Drift 5%	10	CP-norm	0.0000 / 0.000	0.7392	92.1	0.739 / 0.739 / 0.739	2/2/2
LSTM	FD004	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.6928	98.3	0.693 / 0.693 / 0.693	3/3/3
LSTM	FD004	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.6920	98.2	0.692 / 0.692 / 0.692	1/1/1
LSTM	FD004	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.6920	98.2	0.692 / 0.692 / 0.692	2/2/2
LSTM	FD004	Drift 5%	20	Ensemble	0.0000 / 0.000	0.7004	98.5	0.700 / 0.700 / 0.700	4/4/4
LSTM	FD004	Drift 5%	20	CP-norm	0.0000 / 0.000	0.7021	98.5	0.702 / 0.702 / 0.702	5/5/5
LSTM	FD004	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.6557	102.4	0.656 / 0.656 / 0.656	3/3/3
LSTM	FD004	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.6532	102.3	0.653 / 0.653 / 0.653	1/1/1
LSTM	FD004	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.6540	102.3	0.654 / 0.654 / 0.654	2/2/2
LSTM	FD004	Drift 5%	30	Ensemble	0.0000 / 0.000	0.6624	102.6	0.662 / 0.662 / 0.662	5/5/5
LSTM	FD004	Drift 5%	30	CP-norm	0.0000 / 0.000	0.6599	102.5	0.660 / 0.660 / 0.660	4/4/4
Transformer	FD001	Clean	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD001	Clean	10	Fixed var.	0.0000 / 0.000	0.0220	37.0	0.022 / 0.022 / 0.022	5/5/5
Transformer	FD001	Clean	10	MC Drop.	0.0000 / 0.000	0.0200	37.3	0.020 / 0.020 / 0.020	4/4/4
Transformer	FD001	Clean	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD001	Clean	10	CP-norm	0.0000 / 0.000	0.0040	36.0	0.004 / 0.004 / 0.004	3/3/3
Transformer	FD001	Clean	20	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
Transformer	FD001	Clean	20	Fixed var.	0.0000 / 0.000	0.0200	50.6	0.020 / 0.020 / 0.020	4/4/4
Transformer	FD001	Clean	20	MC Drop.	0.0000 / 0.000	0.0260	50.3	0.026 / 0.026 / 0.026	5/5/5
Transformer	FD001	Clean	20	Ensemble	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD001	Clean	20	CP-norm	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	3/3/3
Transformer	FD001	Clean	30	Heterosc.	0.0000 / 0.000	0.0080	54.0	0.008 / 0.008 / 0.008	1/1/1
Transformer	FD001	Clean	30	Fixed var.	0.0000 / 0.000	0.0220	68.0	0.022 / 0.022 / 0.022	3/3/3
Transformer	FD001	Clean	30	MC Drop.	0.0000 / 0.000	0.0240	68.1	0.024 / 0.024 / 0.024	4/4/4
Transformer	FD001	Clean	30	Ensemble	0.0000 / 0.000	0.0100	54.0	0.010 / 0.010 / 0.010	2/2/2
Transformer	FD001	Clean	30	CP-norm	0.0000 / 0.000	0.0380	69.8	0.038 / 0.038 / 0.038	5/5/5
Transformer	FD001	Noise 1%	10	Heterosc.	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD001	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0212	37.1	0.021 / 0.021 / 0.021	5/5/5
Transformer	FD001	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0208	37.6	0.021 / 0.021 / 0.021	4/4/4
Transformer	FD001	Noise 1%	10	Ensemble	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD001	Noise 1%	10	CP-norm	0.0000 / 0.000	0.0040	36.4	0.004 / 0.004 / 0.004	3/3/3
Transformer	FD001	Noise 1%	20	Heterosc.	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD001	Noise 1%	20	Fixed var.	0.0000 / 0.000	0.0208	50.6	0.021 / 0.021 / 0.021	4/4/4
Transformer	FD001	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0260	51.9	0.026 / 0.026 / 0.026	5/5/5
Transformer	FD001	Noise 1%	20	Ensemble	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD001	Noise 1%	20	CP-norm	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	3/3/3
Transformer	FD001	Noise 1%	30	Heterosc.	0.0000 / 0.000	0.0084	54.7	0.008 / 0.008 / 0.008	2/2/2
Transformer	FD001	Noise 1%	30	Fixed var.	0.0000 / 0.000	0.0224	68.6	0.022 / 0.022 / 0.022	3/3/3
Transformer	FD001	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0244	68.9	0.024 / 0.024 / 0.024	4/4/4
Transformer	FD001	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0060	54.0	0.006 / 0.006 / 0.006	1/1/1
Transformer	FD001	Noise 1%	30	CP-norm	0.0000 / 0.000	0.0376	70.0	0.038 / 0.038 / 0.038	5/5/5
Transformer	FD001	Bias 5%	10	Heterosc.	0.0000 / 0.000	0.0024	37.3	0.002 / 0.002 / 0.002	2/2/2
Transformer	FD001	Bias 5%	10	Fixed var.	0.0000 / 0.000	0.0188	37.4	0.019 / 0.019 / 0.019	4/4/4
Transformer	FD001	Bias 5%	10	MC Drop.	0.0000 / 0.000	0.0204	39.1	0.020 / 0.020 / 0.020	5/5/5
Transformer	FD001	Bias 5%	10	Ensemble	0.0000 / 0.000	0.0020	38.0	0.002 / 0.002 / 0.002	1/1/1
Transformer	FD001	Bias 5%	10	CP-norm	0.0000 / 0.000	0.0048	37.0	0.005 / 0.005 / 0.005	3/3/3
Transformer	FD001	Bias 5%	20	Heterosc.	0.0008 / 0.005	0.0000	—	0.004 / 0.016 / 0.080	3/3/5
Transformer	FD001	Bias 5%	20	Fixed var.	0.0000 / 0.000	0.0208	54.2	0.021 / 0.021 / 0.021	4/4/3
Transformer	FD001	Bias 5%	20	MC Drop.	0.0000 / 0.000	0.0236	54.2	0.024 / 0.024 / 0.024	5/5/4
Transformer	FD001	Bias 5%	20	Ensemble	0.0000 / 0.000	0.0000	—	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD001	Bias 5%	20	CP-norm	0.0000 / 0.000	0.0008	65.5	0.001 / 0.001 / 0.001	2/2/2
Transformer	FD001	Bias 5%	30	Heterosc.	0.0004 / 0.002	0.0104	61.8	0.012 / 0.018 / 0.050	2/2/5
Transformer	FD001	Bias 5%	30	Fixed var.	0.0000 / 0.000	0.0236	65.2	0.024 / 0.024 / 0.024	3/3/2
Transformer	FD001	Bias 5%	30	MC Drop.	0.0000 / 0.000	0.0260	65.7	0.026 / 0.026 / 0.026	4/4/3
Transformer	FD001	Bias 5%	30	Ensemble	0.0000 / 0.000	0.0100	60.0	0.010 / 0.010 / 0.010	1/1/1
Transformer	FD001	Bias 5%	30	CP-norm	0.0000 / 0.000	0.0316	70.5	0.032 / 0.032 / 0.032	5/5/4
Transformer	FD001	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.0160	36.9	0.016 / 0.016 / 0.016	1/1/1
Transformer	FD001	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.0680	48.0	0.068 / 0.068 / 0.068	4/4/4
Transformer	FD001	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.0700	48.0	0.070 / 0.070 / 0.070	5/5/5
Transformer	FD001	Drift 5%	10	Ensemble	0.0000 / 0.000	0.0200	36.0	0.020 / 0.020 / 0.020	3/3/3
Transformer	FD001	Drift 5%	10	CP-norm	0.0000 / 0.000	0.0160	36.9	0.016 / 0.016 / 0.016	2/2/2
Transformer	FD001	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.0240	55.5	0.024 / 0.024 / 0.024	2/2/2
Transformer	FD001	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.1020	59.5	0.102 / 0.102 / 0.102	4/4/4
Transformer	FD001	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.1040	60.1	0.104 / 0.104 / 0.104	5/5/5
Transformer	FD001	Drift 5%	20	Ensemble	0.0000 / 0.000	0.0200	58.0	0.020 / 0.020 / 0.020	1/1/1
Transformer	FD001	Drift 5%	20	CP-norm	0.0000 / 0.000	0.0360	65.1	0.036 / 0.036 / 0.036	3/3/3
Transformer	FD001	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.0620	73.4	0.062 / 0.062 / 0.062	1/1/1
Transformer	FD001	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.1320	76.2	0.132 / 0.132 / 0.132	4/4/4
Transformer	FD001	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.1420	76.8	0.142 / 0.142 / 0.142	5/5/5
Transformer	FD001	Drift 5%	30	Ensemble	0.0000 / 0.000	0.1300	73.8	0.130 / 0.130 / 0.130	2/2/2
Transformer	FD001	Drift 5%	30	CP-norm	0.0000 / 0.000	0.1300	84.1	0.130 / 0.130 / 0.130	3/3/3
Transformer	FD002	Clean	10	Heterosc.	0.0000 / 0.000	0.0016	31.0	0.002 / 0.002 / 0.002	1/1/1
Transformer	FD002	Clean	10	Fixed var.	0.0000 / 0.000	0.0466	40.7	0.047 / 0.047 / 0.047	5/5/5
Transformer	FD002	Clean	10	MC Drop.	0.0000 / 0.000	0.0451	40.4	0.045 / 0.045 / 0.045	4/4/4
Transformer	FD002	Clean	10	Ensemble	0.0000 / 0.000	0.0040	31.0	0.004 / 0.004 / 0.004	2/2/2
Transformer	FD002	Clean	10	CP-norm	0.0000 / 0.000	0.0087	39.9	0.009 / 0.009 / 0.009	3/3/3
Transformer	FD002	Clean	20	Heterosc.	0.0008 / 0.004	0.0150	47.7	0.019 / 0.031 / 0.094	1/2/4
Transformer	FD002	Clean	20	Fixed var.	0.0000 / 0.000	0.0640	52.7	0.064 / 0.064 / 0.064	4/4/2
Transformer	FD002	Clean	20	MC Drop.	0.0000 / 0.000	0.0664	52.6	0.066 / 0.066 / 0.066	5/5/3
Transformer	FD002	Clean	20	Ensemble	0.0000 / 0.000	0.0198	45.2	0.020 / 0.020 / 0.020	2/1/1
Transformer	FD002	Clean	20	CP-norm	0.0008 / 0.004	0.0324	53.6	0.036 / 0.048 / 0.111	3/3/5
Transformer	FD002	Clean	30	Heterosc.	0.0000 / 0.000	0.0601	70.9	0.060 / 0.060 / 0.060	1/1/1
Transformer	FD002	Clean	30	Fixed var.	0.0000 / 0.000	0.0632	64.7	0.063 / 0.063 / 0.063	2/2/2
Transformer	FD002	Clean	30	MC Drop.	0.0000 / 0.000	0.0680	65.1	0.068 / 0.068 / 0.068	4/4/4
Transformer	FD002	Clean	30	Ensemble	0.0000 / 0.000	0.0711	66.3	0.071 / 0.071 / 0.071	5/5/5
Transformer	FD002	Clean	30	CP-norm	0.0000 / 0.000	0.0672	70.9	0.067 / 0.067 / 0.067	3/3/3
Transformer	FD002	Noise 1%	10	Heterosc.	0.0008 / 0.009	0.0076	36.2	0.012 / 0.023 / 0.087	2/2/4
Transformer	FD002	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0716	47.8	0.072 / 0.072 / 0.072	5/5/3
Transformer	FD002	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0670	46.3	0.067 / 0.067 / 0.067	4/4/2
Transformer	FD002	Noise 1%	10	Ensemble	0.0000 / 0.000	0.0103	40.2	0.010 / 0.010 / 0.010	1/1/1
Transformer	FD002	Noise 1%	10	CP-norm	0.0011 / 0.013	0.0152	42.2	0.021 / 0.037 / 0.126	3/3/5
Transformer	FD002	Noise 1%	20	Heterosc.	0.0022 / 0.011	0.0311	57.7	0.042 / 0.075 / 0.252	1/2/4

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
Transformer	FD002	Noise 1%	20	Fixed var.	0.0000 / 0.000	0.0840	61.7	0.084 / 0.084 / 0.084	5/4/2
Transformer	FD002	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0832	61.5	0.083 / 0.083 / 0.083	4/3/1
Transformer	FD002	Noise 1%	20	Ensemble	0.0008 / 0.004	0.0395	58.3	0.043 / 0.055 / 0.119	2/1/3
Transformer	FD002	Noise 1%	20	CP-norm	0.0024 / 0.011	0.0474	63.6	0.059 / 0.095 / 0.285	3/5/5
Transformer	FD002	Noise 1%	30	Heterosc.	0.0000 / 0.000	0.0822	78.6	0.082 / 0.082 / 0.082	1/1/1
Transformer	FD002	Noise 1%	30	Fixed var.	0.0000 / 0.000	0.0896	74.6	0.090 / 0.090 / 0.090	2/2/2
Transformer	FD002	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0926	74.7	0.093 / 0.093 / 0.093	4/4/4
Transformer	FD002	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0925	77.6	0.092 / 0.092 / 0.092	3/3/3
Transformer	FD002	Noise 1%	30	CP-norm	0.0002 / 0.001	0.0988	79.3	0.100 / 0.102 / 0.115	5/5/5
Transformer	FD002	Bias 5%	10	Heterosc.	0.0421 / 0.484	0.1513	91.2	0.362 / 0.992 / 4.357	5/5/5
Transformer	FD002	Bias 5%	10	Fixed var.	0.0324 / 0.373	0.1654	87.7	0.327 / 0.814 / 3.406	2/2/2
Transformer	FD002	Bias 5%	10	MC Drop.	0.0340 / 0.391	0.1624	88.3	0.332 / 0.842 / 3.562	3/3/3
Transformer	FD002	Bias 5%	10	Ensemble	0.0269 / 0.309	0.1534	90.8	0.288 / 0.691 / 2.841	1/1/1
Transformer	FD002	Bias 5%	10	CP-norm	0.0406 / 0.467	0.1564	90.4	0.360 / 0.969 / 4.220	4/4/4
Transformer	FD002	Bias 5%	20	Heterosc.	0.0931 / 0.445	0.1443	94.8	0.610 / 2.007 / 9.457	5/5/5
Transformer	FD002	Bias 5%	20	Fixed var.	0.0792 / 0.378	0.1609	91.1	0.557 / 1.745 / 8.082	2/2/2
Transformer	FD002	Bias 5%	20	MC Drop.	0.0809 / 0.386	0.1592	91.4	0.564 / 1.778 / 8.254	3/3/3
Transformer	FD002	Bias 5%	20	Ensemble	0.0625 / 0.298	0.1534	92.1	0.466 / 1.402 / 6.398	1/1/1
Transformer	FD002	Bias 5%	20	CP-norm	0.0854 / 0.408	0.1516	93.7	0.578 / 1.859 / 8.689	4/4/4
Transformer	FD002	Bias 5%	30	Heterosc.	0.0819 / 0.340	0.1456	96.7	0.555 / 1.784 / 8.335	4/5/5
Transformer	FD002	Bias 5%	30	Fixed var.	0.0803 / 0.333	0.1553	95.8	0.557 / 1.762 / 8.187	5/4/4
Transformer	FD002	Bias 5%	30	MC Drop.	0.0794 / 0.329	0.1556	95.7	0.552 / 1.743 / 8.092	3/3/3
Transformer	FD002	Bias 5%	30	Ensemble	0.0538 / 0.223	0.1565	95.2	0.425 / 1.232 / 5.532	1/1/1
Transformer	FD002	Bias 5%	30	CP-norm	0.0757 / 0.314	0.1524	96.7	0.531 / 1.667 / 7.726	2/2/2
Transformer	FD002	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.7265	90.0	0.726 / 0.726 / 0.726	1/1/1
Transformer	FD002	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.7581	91.4	0.758 / 0.758 / 0.758	5/5/5
Transformer	FD002	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.7573	91.3	0.757 / 0.757 / 0.757	4/4/4
Transformer	FD002	Drift 5%	10	Ensemble	0.0000 / 0.000	0.7470	90.9	0.747 / 0.747 / 0.747	3/3/3
Transformer	FD002	Drift 5%	10	CP-norm	0.0000 / 0.000	0.7328	90.3	0.733 / 0.733 / 0.733	2/2/2
Transformer	FD002	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.7012	95.6	0.701 / 0.701 / 0.701	1/1/1
Transformer	FD002	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.7036	95.7	0.704 / 0.704 / 0.704	3/3/3
Transformer	FD002	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.7036	95.7	0.704 / 0.704 / 0.704	4/4/4
Transformer	FD002	Drift 5%	20	Ensemble	0.0000 / 0.000	0.7036	95.7	0.704 / 0.704 / 0.704	5/5/5
Transformer	FD002	Drift 5%	20	CP-norm	0.0000 / 0.000	0.7020	95.7	0.702 / 0.702 / 0.702	2/2/2
Transformer	FD002	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.6522	99.7	0.652 / 0.652 / 0.652	1/1/1
Transformer	FD002	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.6522	99.7	0.652 / 0.652 / 0.652	2/2/2
Transformer	FD002	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.6522	99.7	0.652 / 0.652 / 0.652	3/3/3
Transformer	FD002	Drift 5%	30	Ensemble	0.0000 / 0.000	0.6522	99.7	0.652 / 0.652 / 0.652	4/4/4
Transformer	FD002	Drift 5%	30	CP-norm	0.0000 / 0.000	0.6522	99.7	0.652 / 0.652 / 0.652	5/5/5
Transformer	FD003	Clean	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD003	Clean	10	Fixed var.	0.0000 / 0.000	0.0200	33.0	0.020 / 0.020 / 0.020	4/4/4
Transformer	FD003	Clean	10	MC Drop.	0.0000 / 0.000	0.0200	33.0	0.020 / 0.020 / 0.020	5/5/5
Transformer	FD003	Clean	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD003	Clean	10	CP-norm	0.0000 / 0.000	0.0080	46.8	0.008 / 0.008 / 0.008	3/3/3
Transformer	FD003	Clean	20	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD003	Clean	20	Fixed var.	0.0000 / 0.000	0.0260	46.8	0.026 / 0.026 / 0.026	5/5/5
Transformer	FD003	Clean	20	MC Drop.	0.0000 / 0.000	0.0220	46.7	0.022 / 0.022 / 0.022	4/4/4
Transformer	FD003	Clean	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD003	Clean	20	CP-norm	0.0000 / 0.000	0.0160	69.1	0.016 / 0.016 / 0.016	3/3/3
Transformer	FD003	Clean	30	Heterosc.	0.0000 / 0.000	0.0220	58.2	0.022 / 0.022 / 0.022	1/1/1
Transformer	FD003	Clean	30	Fixed var.	0.0000 / 0.000	0.0380	57.3	0.038 / 0.038 / 0.038	3/3/3
Transformer	FD003	Clean	30	MC Drop.	0.0000 / 0.000	0.0380	57.3	0.038 / 0.038 / 0.038	4/4/4
Transformer	FD003	Clean	30	Ensemble	0.0000 / 0.000	0.0300	60.3	0.030 / 0.030 / 0.030	2/2/2
Transformer	FD003	Clean	30	CP-norm	0.0000 / 0.000	0.0740	62.5	0.074 / 0.074 / 0.074	5/5/5
Transformer	FD003	Noise 1%	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD003	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0200	33.0	0.020 / 0.020 / 0.020	4/4/4
Transformer	FD003	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0200	33.0	0.020 / 0.020 / 0.020	5/5/5
Transformer	FD003	Noise 1%	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD003	Noise 1%	10	CP-norm	0.0000 / 0.000	0.0080	48.0	0.008 / 0.008 / 0.008	3/3/3
Transformer	FD003	Noise 1%	20	Heterosc.	0.0000 / 0.000	0.0004	55.0	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD003	Noise 1%	20	Fixed var.	0.0000 / 0.000	0.0256	46.5	0.026 / 0.026 / 0.026	5/5/5
Transformer	FD003	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0212	47.3	0.021 / 0.021 / 0.021	4/4/4
Transformer	FD003	Noise 1%	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD003	Noise 1%	20	CP-norm	0.0000 / 0.000	0.0148	67.5	0.015 / 0.015 / 0.015	3/3/3
Transformer	FD003	Noise 1%	30	Heterosc.	0.0000 / 0.000	0.0232	59.9	0.023 / 0.023 / 0.023	1/1/1
Transformer	FD003	Noise 1%	30	Fixed var.	0.0000 / 0.000	0.0388	57.2	0.039 / 0.039 / 0.039	4/4/4
Transformer	FD003	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0384	57.2	0.038 / 0.038 / 0.038	3/3/3
Transformer	FD003	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0340	59.7	0.034 / 0.034 / 0.034	2/2/2
Transformer	FD003	Noise 1%	30	CP-norm	0.0000 / 0.000	0.0728	62.8	0.073 / 0.073 / 0.073	5/5/5
Transformer	FD003	Bias 5%	10	Heterosc.	0.0004 / 0.007	0.0048	32.7	0.007 / 0.013 / 0.045	2/2/5
Transformer	FD003	Bias 5%	10	Fixed var.	0.0000 / 0.000	0.0224	35.3	0.022 / 0.022 / 0.022	5/5/4
Transformer	FD003	Bias 5%	10	MC Drop.	0.0000 / 0.000	0.0216	35.3	0.022 / 0.022 / 0.022	4/4/3
Transformer	FD003	Bias 5%	10	Ensemble	0.0000 / 0.000	0.0020	31.0	0.002 / 0.002 / 0.002	1/1/1
Transformer	FD003	Bias 5%	10	CP-norm	0.0000 / 0.000	0.0128	46.5	0.013 / 0.013 / 0.013	3/3/2
Transformer	FD003	Bias 5%	20	Heterosc.	0.0000 / 0.000	0.0052	62.6	0.005 / 0.005 / 0.005	1/1/1

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
Transformer	FD003	Bias 5%	20	Fixed var.	0.0000 / 0.000	0.0524	48.4	0.052 / 0.052 / 0.052	5/5/5
Transformer	FD003	Bias 5%	20	MC Drop.	0.0000 / 0.000	0.0464	48.3	0.046 / 0.046 / 0.046	4/4/4
Transformer	FD003	Bias 5%	20	Ensemble	0.0000 / 0.000	0.0060	53.0	0.006 / 0.006 / 0.006	2/2/2
Transformer	FD003	Bias 5%	20	CP-norm	0.0000 / 0.000	0.0192	63.6	0.019 / 0.019 / 0.019	3/3/3
Transformer	FD003	Bias 5%	30	Heterosc.	0.0000 / 0.000	0.0412	60.4	0.041 / 0.041 / 0.041	1/1/1
Transformer	FD003	Bias 5%	30	Fixed var.	0.0000 / 0.000	0.0520	58.0	0.052 / 0.052 / 0.052	3/3/3
Transformer	FD003	Bias 5%	30	MC Drop.	0.0000 / 0.000	0.0512	57.8	0.051 / 0.051 / 0.051	2/2/2
Transformer	FD003	Bias 5%	30	Ensemble	0.0000 / 0.000	0.0620	58.6	0.062 / 0.062 / 0.062	4/4/4
Transformer	FD003	Bias 5%	30	CP-norm	0.0000 / 0.000	0.0824	65.7	0.082 / 0.082 / 0.082	5/5/5
Transformer	FD003	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.0560	45.2	0.056 / 0.056 / 0.056	1/1/1
Transformer	FD003	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.2140	52.3	0.214 / 0.214 / 0.214	5/5/5
Transformer	FD003	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.2020	51.4	0.202 / 0.202 / 0.202	4/4/4
Transformer	FD003	Drift 5%	10	Ensemble	0.0000 / 0.000	0.0600	44.3	0.060 / 0.060 / 0.060	2/2/2
Transformer	FD003	Drift 5%	10	CP-norm	0.0000 / 0.000	0.0920	48.1	0.092 / 0.092 / 0.092	3/3/3
Transformer	FD003	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.1940	56.2	0.194 / 0.194 / 0.194	1/1/1
Transformer	FD003	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.3680	69.3	0.368 / 0.368 / 0.368	5/5/5
Transformer	FD003	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.3460	66.9	0.346 / 0.346 / 0.346	4/4/4
Transformer	FD003	Drift 5%	20	Ensemble	0.0000 / 0.000	0.2200	57.5	0.220 / 0.220 / 0.220	3/3/3
Transformer	FD003	Drift 5%	20	CP-norm	0.0000 / 0.000	0.2180	59.3	0.218 / 0.218 / 0.218	2/2/2
Transformer	FD003	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.3260	75.4	0.326 / 0.326 / 0.326	1/1/1
Transformer	FD003	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.4580	84.3	0.458 / 0.458 / 0.458	4/4/4
Transformer	FD003	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.4640	84.2	0.464 / 0.464 / 0.464	5/5/5
Transformer	FD003	Drift 5%	30	Ensemble	0.0000 / 0.000	0.4300	81.1	0.430 / 0.430 / 0.430	3/3/3
Transformer	FD003	Drift 5%	30	CP-norm	0.0000 / 0.000	0.3620	78.9	0.362 / 0.362 / 0.362	2/2/2
Transformer	FD004	Clean	10	Heterosc.	0.0000 / 0.000	0.0338	42.4	0.034 / 0.034 / 0.034	1/1/1
Transformer	FD004	Clean	10	Fixed var.	0.0000 / 0.000	0.0582	40.8	0.058 / 0.058 / 0.058	4/4/4
Transformer	FD004	Clean	10	MC Drop.	0.0000 / 0.000	0.0473	40.5	0.047 / 0.047 / 0.047	2/2/2
Transformer	FD004	Clean	10	Ensemble	0.0000 / 0.000	0.0506	44.2	0.051 / 0.051 / 0.051	3/3/3
Transformer	FD004	Clean	10	CP-norm	0.0000 / 0.000	0.0591	44.3	0.059 / 0.059 / 0.059	5/5/5
Transformer	FD004	Clean	20	Heterosc.	0.0000 / 0.000	0.0380	59.2	0.038 / 0.038 / 0.038	1/1/1
Transformer	FD004	Clean	20	Fixed var.	0.0000 / 0.000	0.0489	55.9	0.049 / 0.049 / 0.049	4/4/4
Transformer	FD004	Clean	20	MC Drop.	0.0000 / 0.000	0.0388	54.5	0.039 / 0.039 / 0.039	2/2/2
Transformer	FD004	Clean	20	Ensemble	0.0000 / 0.000	0.0422	59.2	0.042 / 0.042 / 0.042	3/3/3
Transformer	FD004	Clean	20	CP-norm	0.0000 / 0.000	0.0675	62.8	0.068 / 0.068 / 0.068	5/5/5
Transformer	FD004	Clean	30	Heterosc.	0.0000 / 0.000	0.0515	74.1	0.051 / 0.051 / 0.051	4/2/2
Transformer	FD004	Clean	30	Fixed var.	0.0008 / 0.004	0.0464	69.7	0.051 / 0.063 / 0.131	3/4/4
Transformer	FD004	Clean	30	MC Drop.	0.0008 / 0.004	0.0414	69.4	0.046 / 0.058 / 0.126	1/3/3
Transformer	FD004	Clean	30	Ensemble	0.0000 / 0.000	0.0506	70.5	0.051 / 0.051 / 0.051	2/1/1
Transformer	FD004	Clean	30	CP-norm	0.0008 / 0.004	0.0852	78.1	0.089 / 0.102 / 0.170	5/5/5
Transformer	FD004	Noise 1%	10	Heterosc.	0.0003 / 0.006	0.0400	46.0	0.042 / 0.047 / 0.074	1/1/2
Transformer	FD004	Noise 1%	10	Fixed var.	0.0003 / 0.006	0.0751	46.4	0.077 / 0.082 / 0.109	5/5/5
Transformer	FD004	Noise 1%	10	MC Drop.	0.0005 / 0.009	0.0567	43.4	0.059 / 0.067 / 0.107	3/3/4
Transformer	FD004	Noise 1%	10	Ensemble	0.0000 / 0.000	0.0506	47.0	0.051 / 0.051 / 0.051	2/2/1
Transformer	FD004	Noise 1%	10	CP-norm	0.0002 / 0.003	0.0663	50.4	0.067 / 0.070 / 0.083	4/4/3
Transformer	FD004	Noise 1%	20	Heterosc.	0.0000 / 0.000	0.0511	65.5	0.051 / 0.051 / 0.051	1/1/1
Transformer	FD004	Noise 1%	20	Fixed var.	0.0005 / 0.003	0.0657	64.2	0.068 / 0.076 / 0.116	4/4/4
Transformer	FD004	Noise 1%	20	MC Drop.	0.0007 / 0.004	0.0516	59.9	0.055 / 0.065 / 0.119	2/3/5
Transformer	FD004	Noise 1%	20	Ensemble	0.0000 / 0.000	0.0616	65.9	0.062 / 0.062 / 0.062	3/2/2
Transformer	FD004	Noise 1%	20	CP-norm	0.0000 / 0.000	0.0846	69.7	0.085 / 0.085 / 0.085	5/5/3
Transformer	FD004	Noise 1%	30	Heterosc.	0.0010 / 0.005	0.0780	79.2	0.083 / 0.098 / 0.179	2/2/2
Transformer	FD004	Noise 1%	30	Fixed var.	0.0029 / 0.013	0.0717	78.4	0.086 / 0.129 / 0.359	3/4/5
Transformer	FD004	Noise 1%	30	MC Drop.	0.0029 / 0.013	0.0633	76.9	0.078 / 0.121 / 0.350	1/3/4
Transformer	FD004	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0920	80.3	0.092 / 0.092 / 0.092	4/1/1
Transformer	FD004	Noise 1%	30	CP-norm	0.0010 / 0.005	0.1149	83.1	0.120 / 0.135 / 0.216	5/5/3
Transformer	FD004	Bias 5%	10	Heterosc.	0.0133 / 0.226	0.2970	89.9	0.364 / 0.564 / 1.630	2/4/4
Transformer	FD004	Bias 5%	10	Fixed var.	0.0105 / 0.177	0.3180	88.4	0.370 / 0.527 / 1.364	4/2/2
Transformer	FD004	Bias 5%	10	MC Drop.	0.0135 / 0.229	0.3078	88.7	0.375 / 0.578 / 1.658	5/5/5
Transformer	FD004	Bias 5%	10	Ensemble	0.0059 / 0.100	0.3266	91.2	0.356 / 0.445 / 0.917	1/1/1
Transformer	FD004	Bias 5%	10	CP-norm	0.0118 / 0.200	0.3102	90.1	0.369 / 0.546 / 1.492	3/3/3
Transformer	FD004	Bias 5%	20	Heterosc.	0.0366 / 0.217	0.2896	96.8	0.473 / 1.022 / 3.952	5/5/5
Transformer	FD004	Bias 5%	20	Fixed var.	0.0292 / 0.173	0.2992	95.6	0.445 / 0.883 / 3.219	2/2/2
Transformer	FD004	Bias 5%	20	MC Drop.	0.0358 / 0.212	0.2916	96.0	0.471 / 1.007 / 3.870	4/4/4
Transformer	FD004	Bias 5%	20	Ensemble	0.0186 / 0.110	0.3089	97.0	0.402 / 0.680 / 2.165	1/1/1
Transformer	FD004	Bias 5%	20	CP-norm	0.0326 / 0.193	0.2992	96.8	0.462 / 0.951 / 3.557	3/3/3
Transformer	FD004	Bias 5%	30	Heterosc.	0.0417 / 0.186	0.2810	101.1	0.489 / 1.115 / 4.450	5/5/5
Transformer	FD004	Bias 5%	30	Fixed var.	0.0356 / 0.159	0.2873	100.4	0.465 / 0.999 / 3.848	2/2/2
Transformer	FD004	Bias 5%	30	MC Drop.	0.0393 / 0.176	0.2841	100.6	0.481 / 1.071 / 4.217	3/4/4
Transformer	FD004	Bias 5%	30	Ensemble	0.0270 / 0.121	0.2954	101.6	0.430 / 0.835 / 2.996	1/1/1
Transformer	FD004	Bias 5%	30	CP-norm	0.0383 / 0.171	0.2900	101.2	0.482 / 1.056 / 4.121	4/3/3
Transformer	FD004	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.6464	88.7	0.646 / 0.646 / 0.646	1/1/1
Transformer	FD004	Drift 5%	10	Fixed var.	0.0008 / 0.014	0.7519	92.5	0.756 / 0.769 / 0.836	5/5/5
Transformer	FD004	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.7376	91.8	0.738 / 0.738 / 0.738	4/4/4
Transformer	FD004	Drift 5%	10	Ensemble	0.0000 / 0.000	0.7173	91.1	0.717 / 0.717 / 0.717	3/3/3
Transformer	FD004	Drift 5%	10	CP-norm	0.0000 / 0.000	0.6692	90.0	0.669 / 0.669 / 0.669	2/2/2
Transformer	FD004	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.6827	97.9	0.683 / 0.683 / 0.683	2/2/2

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
Transformer	FD004	Drift 5%	20	Fixed var.	0.0008 / 0.005	0.6962	98.5	0.700 / 0.713 / 0.781	4/5/5
Transformer	FD004	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.6911	98.2	0.691 / 0.691 / 0.691	3/3/3
Transformer	FD004	Drift 5%	20	Ensemble	0.0000 / 0.000	0.7046	98.6	0.705 / 0.705 / 0.705	5/4/4
Transformer	FD004	Drift 5%	20	CP-norm	0.0000 / 0.000	0.6793	97.8	0.679 / 0.679 / 0.679	1/1/1
Transformer	FD004	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.6565	102.4	0.657 / 0.657 / 0.657	3/3/3
Transformer	FD004	Drift 5%	30	Fixed var.	0.0008 / 0.004	0.6557	102.4	0.660 / 0.673 / 0.740	4/5/5
Transformer	FD004	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.6532	102.3	0.653 / 0.653 / 0.653	1/1/1
Transformer	FD004	Drift 5%	30	Ensemble	0.0000 / 0.000	0.6624	102.6	0.662 / 0.662 / 0.662	5/4/4
Transformer	FD004	Drift 5%	30	CP-norm	0.0000 / 0.000	0.6540	102.4	0.654 / 0.654 / 0.654	2/2/2

S2B. RANK REVERSAL ACROSS COST RATIOS

Reversal: the ranking of the five mechanisms differs between at least two of the cost ratios; top change: the top-ranked mechanism differs. Totals: 34 of 96 reversals, 15 top changes.

Backbone	Data	Cond.	L	Reversal	Top change
LSTM	FD001	Clean	10	no	no
LSTM	FD001	Clean	20	yes	no
LSTM	FD001	Clean	30	no	no
LSTM	FD001	Noise 1%	10	no	no
LSTM	FD001	Noise 1%	20	yes	no
LSTM	FD001	Noise 1%	30	yes	no
LSTM	FD001	Bias 5%	10	no	no
LSTM	FD001	Bias 5%	20	yes	no
LSTM	FD001	Bias 5%	30	yes	no
LSTM	FD001	Drift 5%	10	no	no
LSTM	FD001	Drift 5%	20	no	no
LSTM	FD001	Drift 5%	30	no	no
LSTM	FD002	Clean	10	no	no
LSTM	FD002	Clean	20	no	no
LSTM	FD002	Clean	30	no	no
LSTM	FD002	Noise 1%	10	yes	yes
LSTM	FD002	Noise 1%	20	yes	yes
LSTM	FD002	Noise 1%	30	yes	no
LSTM	FD002	Bias 5%	10	no	no
LSTM	FD002	Bias 5%	20	no	no
LSTM	FD002	Bias 5%	30	no	no
LSTM	FD002	Drift 5%	10	no	no
LSTM	FD002	Drift 5%	20	no	no
LSTM	FD002	Drift 5%	30	no	no
LSTM	FD003	Clean	10	no	no
LSTM	FD003	Clean	20	no	no
LSTM	FD003	Clean	30	no	no
LSTM	FD003	Noise 1%	10	no	no
LSTM	FD003	Noise 1%	20	no	no
LSTM	FD003	Noise 1%	30	no	no
LSTM	FD003	Bias 5%	10	no	no
LSTM	FD003	Bias 5%	20	yes	no
LSTM	FD003	Bias 5%	30	yes	yes
LSTM	FD003	Drift 5%	10	no	no
LSTM	FD003	Drift 5%	20	no	no
LSTM	FD003	Drift 5%	30	no	no
LSTM	FD004	Clean	10	yes	yes
LSTM	FD004	Clean	20	yes	yes
LSTM	FD004	Clean	30	yes	yes
LSTM	FD004	Noise 1%	10	yes	yes
LSTM	FD004	Noise 1%	20	yes	yes
LSTM	FD004	Noise 1%	30	yes	yes
LSTM	FD004	Bias 5%	10	yes	yes
LSTM	FD004	Bias 5%	20	yes	no
LSTM	FD004	Bias 5%	30	yes	no
LSTM	FD004	Drift 5%	10	no	no
LSTM	FD004	Drift 5%	20	no	no
LSTM	FD004	Drift 5%	30	no	no
Transformer	FD001	Clean	10	no	no
Transformer	FD001	Clean	20	no	no
Transformer	FD001	Clean	30	no	no
Transformer	FD001	Noise 1%	10	no	no
Transformer	FD001	Noise 1%	20	no	no

Backbone	Data	Cond.	L	Reversal	Top change
Transformer	FD001	Noise 1%	30	no	no
Transformer	FD001	Bias 5%	10	no	no
Transformer	FD001	Bias 5%	20	yes	no
Transformer	FD001	Bias 5%	30	yes	no
Transformer	FD001	Drift 5%	10	no	no
Transformer	FD001	Drift 5%	20	no	no
Transformer	FD001	Drift 5%	30	no	no
Transformer	FD002	Clean	10	no	no
Transformer	FD002	Clean	20	yes	yes
Transformer	FD002	Clean	30	no	no
Transformer	FD002	Noise 1%	10	yes	no
Transformer	FD002	Noise 1%	20	yes	yes
Transformer	FD002	Noise 1%	30	no	no
Transformer	FD002	Bias 5%	10	no	no
Transformer	FD002	Bias 5%	20	no	no
Transformer	FD002	Bias 5%	30	yes	no
Transformer	FD002	Drift 5%	10	no	no
Transformer	FD002	Drift 5%	20	no	no
Transformer	FD002	Drift 5%	30	no	no
Transformer	FD003	Clean	10	no	no
Transformer	FD003	Clean	20	no	no
Transformer	FD003	Clean	30	no	no
Transformer	FD003	Noise 1%	10	no	no
Transformer	FD003	Noise 1%	20	no	no
Transformer	FD003	Noise 1%	30	no	no
Transformer	FD003	Bias 5%	10	yes	no
Transformer	FD003	Bias 5%	20	no	no
Transformer	FD003	Bias 5%	30	no	no
Transformer	FD003	Drift 5%	10	no	no
Transformer	FD003	Drift 5%	20	no	no
Transformer	FD003	Drift 5%	30	no	no
Transformer	FD004	Clean	10	no	no
Transformer	FD004	Clean	20	no	no
Transformer	FD004	Clean	30	yes	yes
Transformer	FD004	Noise 1%	10	yes	yes
Transformer	FD004	Noise 1%	20	yes	no
Transformer	FD004	Noise 1%	30	yes	yes
Transformer	FD004	Bias 5%	10	yes	no
Transformer	FD004	Bias 5%	20	no	no
Transformer	FD004	Bias 5%	30	yes	no
Transformer	FD004	Drift 5%	10	no	no
Transformer	FD004	Drift 5%	20	yes	no
Transformer	FD004	Drift 5%	30	yes	no

S2C. PER-ENGINE BOOTSTRAP OF TOP-RANKED MECHANISM AGAINST THE OTHERS

For each (condition, L , ratio) the cost difference between the top-ranked mechanism and each of the other four was bootstrapped over engines (2000 resamples). Number of the 144 comparisons per backbone–dataset whose 95% interval excludes zero:

Backbone	Data	Excluding zero
LSTM	FD001	79 / 144
LSTM	FD002	81 / 144
LSTM	FD003	104 / 144
LSTM	FD004	65 / 144
Transformer	FD001	63 / 144
Transformer	FD002	90 / 144
Transformer	FD003	87 / 144
Transformer	FD004	95 / 144

The full 1152 intervals are in the source file.

S3. SINGLE-LEVEL WEIGHTED INTERVAL SCORE

Sources: `leakfree_r2/interval_score_wis_clean.json` (heteroscedastic, ensemble, CP-norm) and `leakfree_r2/fair_calibration_main_table.json` (fixed variance and MC Dropout, calibration-set scale).

With one interval level ($\alpha = 0.10$, $K = 1$) the score used is $\text{WIS} = \frac{1}{K+1/2} \left[\frac{1}{2} |y - \hat{\mu}| + \frac{\alpha}{2} \text{IS}_\alpha \right]$, with $\hat{\mu}$ in place of the median. Means over the terminal unit, clean inputs, five seeds (ensemble: single $M = 5$).

Backbone	Data	Heterosc.	Fixed var.	MC Drop.	Ensemble	CP-norm
LSTM	FD001	5.92	5.97	5.94	5.59	5.90
LSTM	FD002	6.15	6.79	6.82	5.81	6.16
LSTM	FD003	5.20	5.64	5.59	4.76	5.21
LSTM	FD004	7.54	7.55	7.54	7.23	7.53
Transformer	FD001	5.60	5.56	5.31	5.22	5.56
Transformer	FD002	6.14	6.40	6.43	5.79	6.15
Transformer	FD003	5.40	5.59	5.22	4.83	5.46
Transformer	FD004	7.16	7.23	7.20	6.61	7.16

S4. LATENCY DISTRIBUTIONS

Source: leakfree_r2/latency_redesign_run1.json. Per-sample latency in microseconds, median (IQR) over eleven repetitions, one NVIDIA RTX A5000 Laptop GPU; no repetition excluded. *CUDA-event timing*

Data	Backbone	Batch	Heterosc.	Fixed var.	CP-norm	MC Drop. ($T=50$)	Ensemble ($M=5$)
FD001	LSTM	1	553.89 (71.65)	402.37 (24.66)	527.46 (29.74)	13801.47 (1039.97)	1761.28 (91.54)
FD001	LSTM	32	10.98 (1.04)	9.69 (0.32)	10.56 (0.97)	436.10 (16.21)	44.87 (1.34)
FD001	LSTM	512	1.80 (0.12)	1.68 (0.08)	1.72 (0.08)	89.64 (1.10)	8.02 (0.22)
FD001	Transformer	1	802.82 (67.09)	717.82 (37.25)	798.66 (43.89)	48316.42 (1233.41)	3516.42 (57.34)
FD001	Transformer	32	21.98 (1.88)	20.61 (1.22)	23.20 (1.21)	1940.29 (64.85)	113.95 (3.30)
FD001	Transformer	512	5.22 (0.09)	5.69 (0.11)	5.26 (0.05)	420.23 (0.40)	35.33 (0.18)
FD002	LSTM	1	364.54 (46.51)	324.54 (32.78)	366.59 (28.05)	15504.38 (355.22)	1605.63 (78.40)
FD002	LSTM	32	12.77 (3.09)	10.50 (1.09)	11.93 (0.61)	498.24 (11.05)	56.42 (1.86)
FD002	LSTM	512	2.80 (0.06)	2.73 (0.04)	2.78 (0.07)	140.70 (0.81)	13.23 (0.07)
FD002	Transformer	1	827.39 (76.24)	761.86 (87.02)	812.03 (98.37)	58784.77 (1043.97)	3904.51 (136.19)
FD002	Transformer	32	25.63 (2.13)	24.96 (2.18)	25.38 (1.40)	2109.92 (82.58)	127.04 (3.76)
FD002	Transformer	512	27.94 (1.07)	25.65 (1.24)	28.07 (0.54)	442.88 (40.97)	36.61 (0.04)
FD003	LSTM	1	351.23 (13.31)	318.46 (20.48)	335.87 (29.70)	14606.34 (1037.82)	1685.50 (80.42)
FD003	LSTM	32	12.13 (1.17)	9.50 (0.44)	11.55 (0.58)	487.29 (10.01)	55.74 (3.15)
FD003	LSTM	512	2.81 (0.04)	2.80 (0.05)	2.90 (0.06)	140.14 (0.48)	13.07 (0.20)
FD003	Transformer	1	755.71 (126.98)	681.09 (123.94)	739.33 (58.82)	57734.14 (2224.64)	3513.34 (39.94)
FD003	Transformer	32	25.32 (1.04)	22.21 (1.38)	25.38 (1.14)	1828.22 (58.99)	105.38 (3.37)
FD003	Transformer	512	7.06 (0.05)	7.04 (0.05)	7.16 (0.11)	418.41 (0.56)	35.34 (0.34)
FD004	LSTM	1	355.33 (24.03)	282.62 (9.30)	347.14 (13.79)	15542.27 (262.64)	1697.79 (80.74)
FD004	LSTM	32	11.07 (1.14)	8.61 (0.64)	10.78 (1.15)	493.25 (17.49)	51.46 (2.17)
FD004	LSTM	512	2.82 (0.08)	2.79 (0.08)	2.80 (0.08)	139.76 (0.96)	12.97 (0.14)
FD004	Transformer	1	887.81 (86.02)	786.43 (84.18)	798.69 (46.51)	57554.94 (26801.15)	4062.21 (237.57)
FD004	Transformer	32	25.25 (2.43)	20.49 (0.99)	22.18 (0.42)	2005.98 (124.10)	117.09 (3.34)
FD004	Transformer	512	6.15 (0.11)	6.34 (0.29)	6.27 (0.17)	419.99 (0.38)	35.56 (0.10)

Wall-clock timing

Data	Backbone	Batch	Heterosc.	Fixed var.	CP-norm	MC Drop. ($T=50$)	Ensemble ($M=5$)
FD001	LSTM	1	500.51 (24.31)	359.83 (38.64)	542.25 (54.12)	15656.49 (786.87)	1717.93 (124.91)
FD001	LSTM	32	10.41 (0.58)	8.72 (0.68)	10.06 (0.33)	435.96 (11.28)	43.46 (1.91)
FD001	LSTM	512	1.74 (0.05)	1.70 (0.03)	1.74 (0.02)	87.15 (1.46)	7.94 (0.16)
FD001	Transformer	1	749.24 (47.45)	699.36 (13.40)	747.39 (22.79)	52755.86 (3971.31)	3498.63 (82.77)
FD001	Transformer	32	22.17 (2.00)	21.08 (1.94)	21.70 (1.11)	2063.85 (139.81)	110.71 (4.05)
FD001	Transformer	512	5.22 (0.13)	5.75 (0.02)	5.36 (0.06)	420.83 (0.70)	35.46 (0.10)
FD002	LSTM	1	368.77 (15.24)	303.89 (20.86)	360.94 (35.11)	15178.14 (1757.13)	1603.89 (48.64)
FD002	LSTM	32	12.20 (0.66)	10.30 (1.23)	12.05 (0.53)	503.28 (7.08)	57.77 (1.51)
FD002	LSTM	512	2.90 (0.01)	2.86 (0.02)	2.90 (0.02)	143.09 (1.22)	13.40 (0.05)
FD002	Transformer	1	857.85 (77.62)	739.60 (37.45)	839.28 (75.11)	55607.90 (3107.32)	3703.63 (223.77)
FD002	Transformer	32	25.37 (1.80)	24.21 (3.85)	24.88 (2.09)	2053.94 (259.80)	126.65 (2.93)
FD002	Transformer	512	29.50 (0.76)	29.11 (2.66)	29.59 (1.25)	433.07 (2.95)	36.69 (0.32)
FD003	LSTM	1	359.95 (13.19)	333.89 (14.81)	361.56 (22.88)	15153.77 (963.71)	1732.46 (55.40)
FD003	LSTM	32	11.26 (0.70)	9.20 (0.57)	12.08 (7.44)	487.54 (20.19)	59.35 (3.80)
FD003	LSTM	512	2.87 (0.05)	2.91 (0.09)	3.01 (0.14)	140.99 (6.00)	12.57 (0.09)
FD003	Transformer	1	702.29 (54.09)	619.36 (25.58)	691.11 (46.38)	57821.99 (1640.56)	3497.32 (82.45)
FD003	Transformer	32	23.94 (0.67)	21.36 (1.88)	24.68 (2.19)	1833.62 (120.30)	108.21 (7.14)
FD003	Transformer	512	7.22 (0.05)	7.18 (0.05)	7.20 (0.06)	417.44 (0.40)	35.69 (0.60)
FD004	LSTM	1	311.82 (26.08)	268.99 (30.87)	324.69 (34.08)	14728.66 (802.19)	1643.36 (38.18)
FD004	LSTM	32	9.78 (0.90)	8.66 (0.44)	9.74 (0.33)	512.11 (16.48)	53.01 (3.46)
FD004	LSTM	512	2.89 (0.09)	2.88 (0.10)	2.82 (0.13)	137.89 (0.87)	12.99 (0.21)
FD004	Transformer	1	783.19 (61.40)	705.95 (50.81)	791.37 (48.03)	54753.97 (2731.39)	4268.16 (406.60)
FD004	Transformer	32	22.22 (0.97)	21.35 (2.48)	21.25 (1.22)	2047.53 (114.06)	116.07 (1.92)
FD004	Transformer	512	6.14 (0.04)	7.02 (0.14)	6.33 (0.16)	412.15 (0.27)	35.12 (0.15)